

# Lecture 14: The Dimension (continued)

Instructors:

Christoph Lohrmann ([christophl@ru.is](mailto:christophl@ru.is))

*[Based on slides for „Decoding the Matrix“]*



WEEK

SEVEN



# Agenda

## Chapter 6

### Lecture 14:

Direct Sum  
Linear Function Invertibility  
Extracting an invertible Function  
Rank Nullity  
Matrix Invertibility  
Representation of Vector Spaces  
Annihilator





# Direct Sum (I)

Let  $\mathcal{U}$  and  $\mathcal{V}$  be two vector spaces consisting of  $D$ -vectors over a field  $\mathbb{F}$ .

## Condition!

**Definition:** If  $\mathcal{U}$  and  $\mathcal{V}$  share only the zero vector then we define the *direct sum* of  $\mathcal{U}$  and  $\mathcal{V}$  to be the set

$$\{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{U}, \mathbf{v} \in \mathcal{V}\}$$

written  $\mathcal{U} \oplus \mathcal{V}$

That is,  $\mathcal{U} \oplus \mathcal{V}$  is the set of all sums of a vector in  $\mathcal{U}$  and a vector in  $\mathcal{V}$ .



# Direct Sum (II)

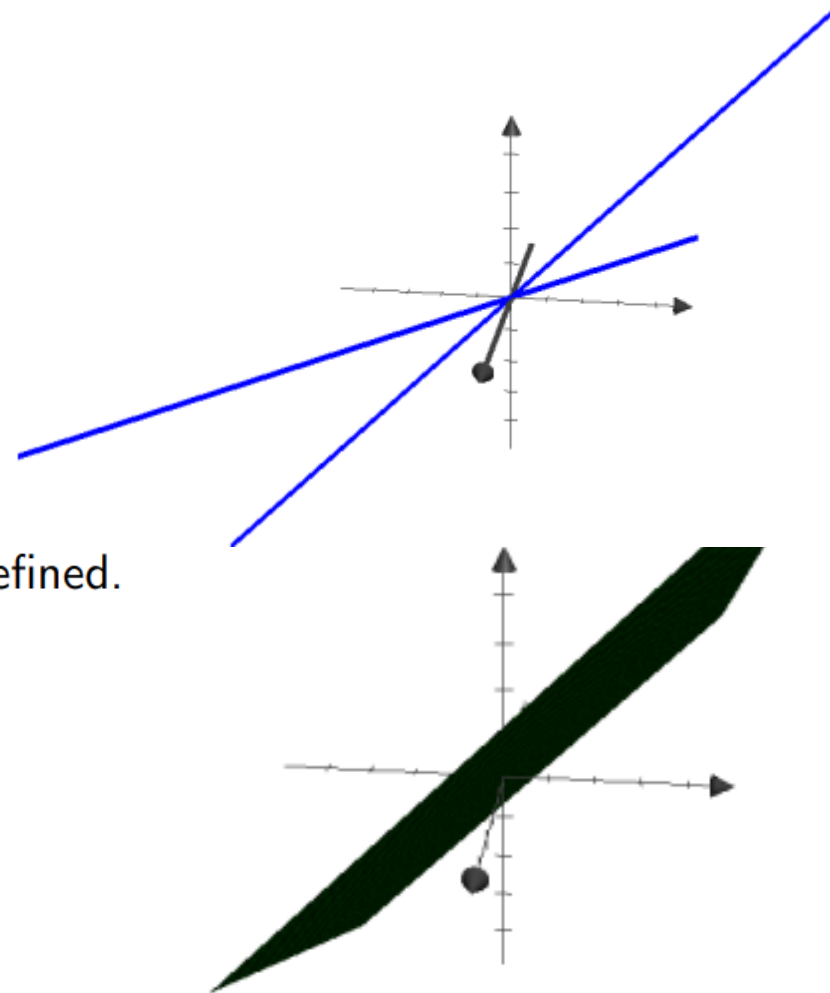
Vectors over  $\mathbb{R}$ :

**Example:**

- ▶ Let  $\mathcal{U} = \text{Span} \{[4, -1, 1]\}$ . (Line)
- ▶ Let  $\mathcal{V} = \text{Span} \{[0, 1, 1]\}$ . (Line)

The only intersection is at the origin, so  $\mathcal{U} \oplus \mathcal{V}$  is defined.

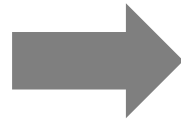
- ▶  $\mathcal{U} \oplus \mathcal{V}$  is the set of vectors  $\mathbf{u} + \mathbf{v}$  where  $\mathbf{u} \in \mathcal{U}$  and  $\mathbf{v} \in \mathcal{V}$ .
- ▶ This is just  $\text{Span} \{[4, -1, 1], [0, 1, 1]\}$  (Plane)
- ▶ Plane containing the two lines





# Direct Sum (III)

- ▶ Let  $\mathcal{U} = \text{Span} \{[4, -1, 1]\}$ .
- ▶ Let  $\mathcal{V} = \text{Span} \{[0, 1, 1]\}$ .



$\mathcal{U} \oplus \mathcal{V} = \text{Span} \{[4, -1, 1], [0, 1, 1]\}$  (Plane)

Vector(s) in  $\mathcal{U}$  (blue arrow pointing to  $[4, -1, 1]$ )  
 Vector(s) in  $\mathcal{V}$  (red arrow pointing to  $[0, 1, 1]$ )

The direct sum of  $\mathcal{U}$  and  $\mathcal{V}$  spans a larger vector space  $\rightarrow$   
 two original vector spaces are subspaces of the new vector space  $\mathcal{U} \oplus \mathcal{V}$



# Direct Sum (III)

Vectors over  $\mathbb{R}$ :

**Example:** Let  $\mathcal{U} = \text{Span} \{[1, 2, 1, 2], [3, 0, 0, 4]\}$  and let  $\mathcal{V} = \{[2, -2, -1, 2]\}$

$$\alpha_1 [1, 2, 1, 2] + \alpha_2 [3, 0, 0, 4] \quad \text{If } \alpha_1 = 0 \text{ and } \alpha_2 = 0 \quad 0[1, 2, 1, 2] + 0[3, 0, 0, 4] = [0, 0, 0, 0]$$

$$\beta_1 [2, -2, -1, 2] \quad \text{If } \beta_1 = 0 \quad 0[2, -2, -1, 2] = [0, 0, 0, 0]$$

They share the zero vector



Is there  $\alpha_1$  and  $\alpha_2$  so that:  $\alpha_1 [1, 2, 1, 2] + \alpha_2 [3, 0, 0, 4] = [2, -2, -1, 2]$  Yes,  $\alpha_1 = -1$  and  $\alpha_2 = 1$

The vector  $[2, -2, -1, 2]$  is in  $\mathcal{U}$  because it is  $[3, 0, 0, 4] - [1, 2, 1, 2]$  

Therefore we cannot form  $\mathcal{U} \oplus \mathcal{V}$ .

**They do not share ONLY the zero vector**  
(blue is in span red, so everything in span blue is also in span red)



# Direct Sum (IV)

For  $GF(2)$  we can even calculate all vectors in the new span



Vectors over  $GF(2)$ :

**Example:** Let  $\mathcal{U} = \text{Span} \{1000, 0100\}$  and let  $\mathcal{V} = \text{Span} \{0010\}$ .

It is fairly obvious that  $\mathcal{U}$  and  $\mathcal{V}$  only share the zero vector (but can also be easily proven by showing that the vector in  $\mathcal{V}$  is not in the span of the vectors in  $\mathcal{U}$ ).

Therefore  $\mathcal{U} \oplus \mathcal{V}$  is defined.

$$\mathcal{U} \oplus \mathcal{V} = \text{Span}\{[1,0,0,0], [0,1,0,0], [0,1,0,0]\}$$



# Direct Sum ( $\mathcal{V}$ )

For GF(2) we can even calculate all vectors in the new span



$$\mathcal{U} \oplus \mathcal{V} = \{ \text{0000} + \text{0000}, \text{1000} + \text{0000}, \text{0100} + \text{0000}, \text{1100} + \text{0000}, \text{0000} + \text{0010}, \text{1000} + \text{0010}, \text{0100} + \text{0010}, \text{1100} + \text{0010} \}$$

which is equal to  $\{ \text{0000}, \text{1000}, \text{0100}, \text{1100}, \text{0010}, \text{1010}, \text{0110}, \text{1110} \}$ .

Original (incl. Zero Vector)

Sum (new)



# Direct Sum - Properties (I)

## Part 1

**Lemma:**  $\mathcal{U} \oplus \mathcal{V}$  is a vector space.  
(Prove using Properties V1, V2, V3.)

## Part 2

**Lemma:** The union of

- ▶ a set of generators of  $\mathcal{U}$ , and
- ▶ a set of generators of  $\mathcal{V}$

is a set of generators for  $\mathcal{U} \oplus \mathcal{V}$ .

← **Direct Sum**

Remember (some lectures ago)

**Property V1** Subset contains the zero vector  $\mathbf{0}$

**Property V2** If subset contains  $\mathbf{v}$  then it contains  $\alpha \mathbf{v}$  for every scalar  $\alpha$

**Property V3** If subset contains  $\mathbf{u}$  and  $\mathbf{v}$  then it contains  $\mathbf{u} + \mathbf{v}$

**Proof:** Suppose  $\mathcal{U} = \text{Span} \{\mathbf{u}_1, \dots, \mathbf{u}_m\}$  and  $\mathcal{V} = \text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ .  
Then

- ▶ every vector in  $\mathcal{U}$  can be written as  $\alpha_1 \mathbf{u}_1 + \dots + \alpha_m \mathbf{u}_m$ , and
- ▶ every vector in  $\mathcal{V}$  can be written as  $\beta_1 \mathbf{v}_1 + \dots + \beta_n \mathbf{v}_n$

so every vector in  $\mathcal{U} \oplus \mathcal{V}$  can be written as

$$\alpha_1 \mathbf{u}_1 + \dots + \alpha_m \mathbf{u}_m + \beta_1 \mathbf{v}_1 + \dots + \beta_n \mathbf{v}_n$$

**QED**



# Direct Sum - Properties (II)

## Part 2

### Direct Sum Basis Lemma:

Union of a basis of  $\mathcal{U}$  and a basis of  $\mathcal{V}$  is a basis of  $\mathcal{U} \oplus \mathcal{V}$ .



**Proof:** Clearly

- ▶ a basis of  $\mathcal{U}$  is a set of generators for  $\mathcal{U}$ , and
- ▶ a basis of  $\mathcal{V}$  is a set of generators for  $\mathcal{V}$ .

## Part 2

Therefore the previous lemma shows that

- ▶ the union of a basis for  $\mathcal{U}$  and a basis for  $\mathcal{V}$  is a generating set for  $\mathcal{U} \oplus \mathcal{V}$ .

We just need to show that the union is linearly independent. **(to be continued)**



# Direct Sum - Properties (III)



**Proof, cont'd:** Let  $\{\mathbf{u}_1, \dots, \mathbf{u}_m\}$  be a basis for  $\mathcal{U}$ . Let  $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$  be a basis for  $\mathcal{V}$ .

We need to show that  $\{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n\}$  is independent.

Suppose

$$\mathbf{0} = \alpha_1 \mathbf{u}_1 + \dots + \alpha_m \mathbf{u}_m + \beta_1 \mathbf{v}_1 + \dots + \beta_n \mathbf{v}_n.$$

Then

$$\underbrace{\alpha_1 \mathbf{u}_1 + \dots + \alpha_m \mathbf{u}_m}_{\text{in } \mathcal{U}} = \underbrace{(-\beta_1) \mathbf{v}_1 + \dots + (-\beta_n) \mathbf{v}_n}_{\text{in } \mathcal{V}}$$

Left-hand side is a vector in  $\mathcal{U}$ , and right-hand side is a vector in  $\mathcal{V}$ .

By definition of  $\mathcal{U} \oplus \mathcal{V}$ , the only vector in both  $\mathcal{U}$  and  $\mathcal{V}$  is the zero vector. This shows:

$$\mathbf{0} = \alpha_1 \mathbf{u}_1 + \dots + \alpha_m \mathbf{u}_m$$

and

$$\mathbf{0} = (-\beta_1) \mathbf{v}_1 + \dots + (-\beta_n) \mathbf{v}_n$$

By linear independence, the linear combinations must be trivial.

(since we know each is a basis by itself)

One cannot be represented  
via a non-trivial linear  
combination of the other

**QED**



# Direct Sum - Dimension

## Direct Sum Basis Lemma:

Union of a basis of  $\mathcal{U}$  and a basis of  $\mathcal{V}$  is a basis of  $\mathcal{U} \oplus \mathcal{V}$ .

**Direct Sum Dimension Corollary:**  $\dim \mathcal{U} + \dim \mathcal{V} = \dim \mathcal{U} \oplus \mathcal{V}$

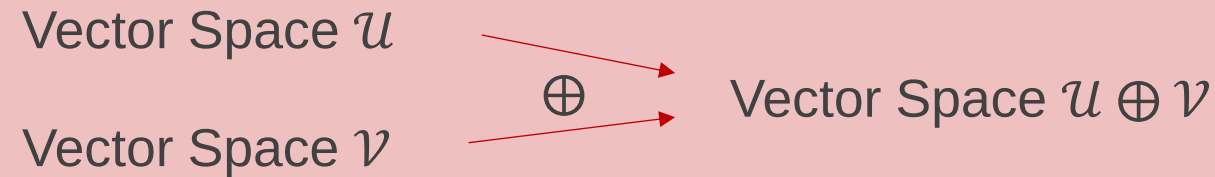


**Proof:** A basis for  $\mathcal{U}$  together with a basis for  $\mathcal{V}$  forms a basis for  $\mathcal{U} \oplus \mathcal{V}$ . QED

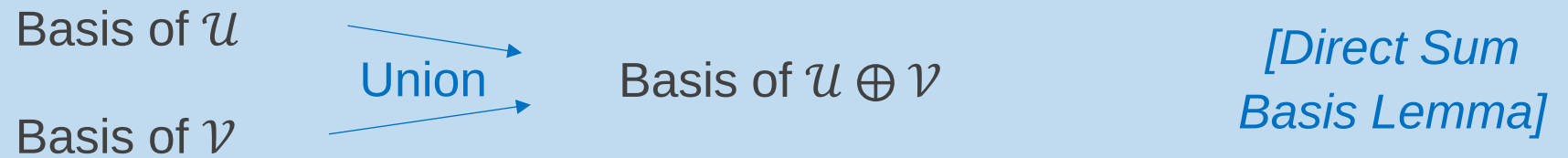


# Direct Sum - Summary

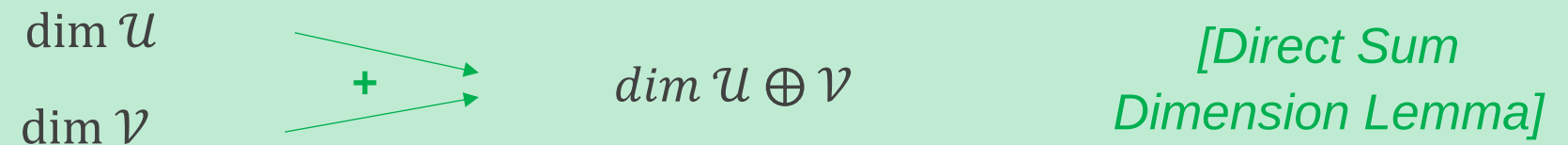
**Vector Space:**



**Basis:**



**Dimension:**





# Linear Function Invertibility (I)

How to tell if a linear function  $f : \mathcal{V} \longrightarrow \mathcal{W}$  is invertible?

- ▶ **One-to-one?**  $f$  is one-to-one if its kernel is trivial. Equivalent: if its kernel has dimension zero. (Only Zero vector)



- ▶ **Onto?**  $f$  is onto if its image equals its co-domain





# Linear Function Invertibility (II) $f : \mathcal{V} \longrightarrow \mathcal{W}$

**Dimension Lemma:** If  $\mathcal{U}$  is a subspace of  $\mathcal{W}$  then

Property D1:  $\dim \mathcal{U} \leq \dim \mathcal{W}$ , and

Property D2: if  $\dim \mathcal{U} = \dim \mathcal{W}$  then  $\mathcal{U} = \mathcal{W}$

Remember  
(last lecture)

$f$  is invertible iff  $\dim \text{Ker } f = 0$  and  $\dim \text{Im } f = \dim \mathcal{W}$

**One-to-one**

**Onto**

(Dimension is the Size of a  
Basis for a Vector Space)

need  $\dim \mathcal{V} = \dim \mathcal{W}$ .

If these dimensions are the same, these spaces are the same!





largest invertible subfunction

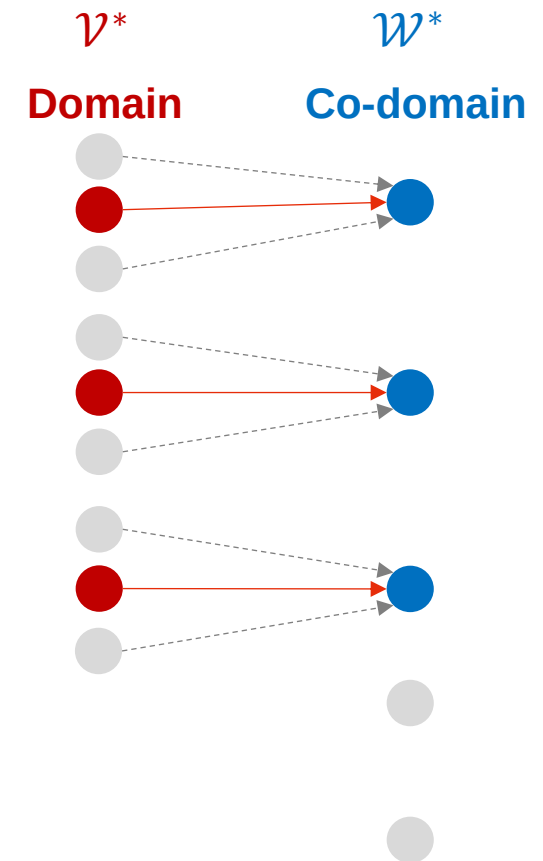
# Extracting an invertible Function (III)

- Choose smaller co-domain  $\mathcal{W}^*$   
Let  $\mathcal{W}^*$  be image of  $f$

Let  $\mathbf{w}_1, \dots, \mathbf{w}_r$  be a basis of  $\mathcal{W}^*$

We select  $w_1, \dots, w_r$  so that they are a basis

⋮





largest invertible subfunction

# Extracting an invertible Function (IV)

- Choose smaller co-domain  $\mathcal{W}^*$   
Let  $\mathcal{W}^*$  be image of  $f$

Let  $\underline{\mathbf{w}_1, \dots, \mathbf{w}_r}$  be a basis of  $\mathcal{W}^*$

We select  $w_1, \dots, w_r$  so that they are a basis

- Choose smaller domain  $\mathcal{V}^*$   
Let  $\underline{\mathbf{v}_1, \dots, \mathbf{v}_r}$  be pre-images of  $\mathbf{w}_1, \dots, \mathbf{w}_r$

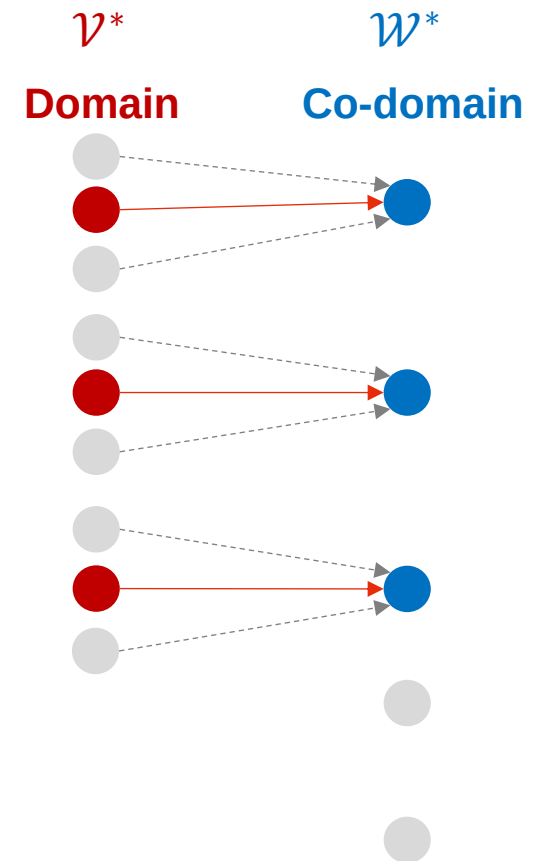
We don't know yet, if they are a basis, too

That is,  $f(\mathbf{v}_1) = \mathbf{w}_1, \dots, f(\mathbf{v}_r) = \mathbf{w}_r$

Let  $\mathcal{V}^* = \text{Span} \{ \mathbf{v}_1, \dots, \mathbf{v}_r \}$

- Define function  $f^* : \mathcal{V}^* \longrightarrow \mathcal{W}^*$   
by  $f^*(\mathbf{x}) = f(\mathbf{x})$

Important to remember for later proofs – these are input-output pairs





largest invertible subfunction

# Extracting an invertible Function (VI)

We will show

- ▶  $f^*$  is onto
- ▶  $f^*$  is one-to-one (kernel is trivial)

## Onto:

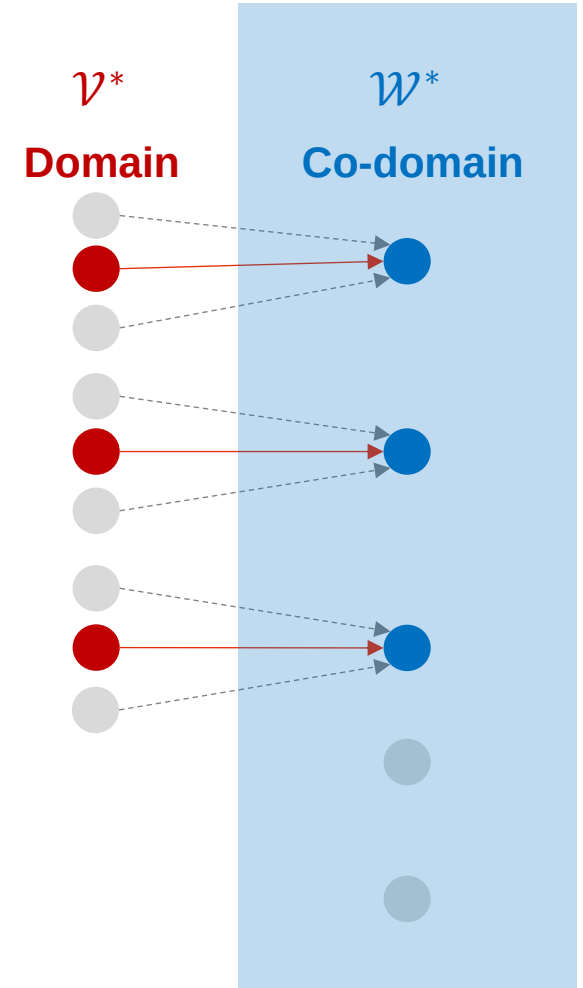
Let  $\mathbf{w}$  be any vector in co-domain  $\mathcal{W}^*$ .  
There are scalars  $\alpha_1, \dots, \alpha_r$  such that

$$\mathbf{w} = \alpha_1 \mathbf{w}_1 + \dots + \alpha_r \mathbf{w}_r$$

Because  $f$  is linear,

$$\begin{aligned} f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r) &= \alpha_1 f(\mathbf{v}_1) + \dots + \alpha_r f(\mathbf{v}_r) \quad (\text{Remember previous slide}) \\ &= \alpha_1 \mathbf{w}_1 + \dots + \alpha_r \mathbf{w}_r \end{aligned}$$

so  $\mathbf{w}$  is image of  $\alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r \in \mathcal{V}^*$  **QED**





largest invertible subfunction

# Extracting an invertible Function (VII)



## One-to-one:

By **One-to-One Lemma**, need only show kernel is trivial.

Suppose  $\mathbf{v}^*$  is in  $\mathcal{V}^*$  and  $f(\mathbf{v}^*) = \mathbf{0}$  (Kernel)

Because  $\mathcal{V}^* = \text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ , there are scalars  $\alpha_1, \dots, \alpha_r$  such that

$$\mathbf{v}^* = \alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r$$

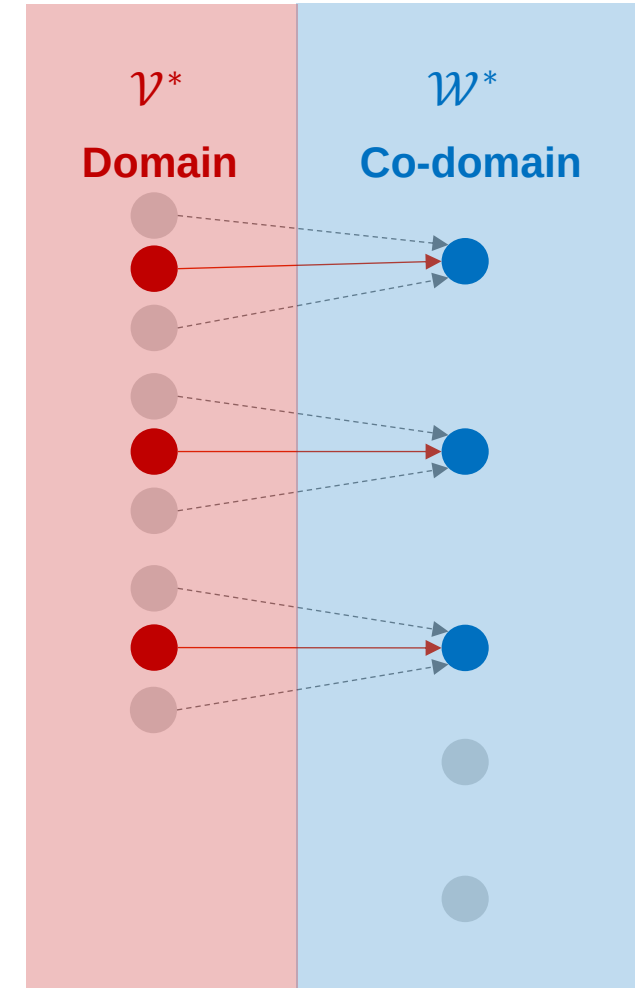
Applying  $f$  to both sides,

$$\begin{aligned} \mathbf{0} &= f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r) \\ &= \alpha_1 \mathbf{w}_1 + \dots + \alpha_r \mathbf{w}_r \end{aligned}$$

Because  $\mathbf{w}_1, \dots, \mathbf{w}_r$  are linearly independent,  $\alpha_1 = \dots = \alpha_r = 0$  Only the trivial linear combination

so  $\mathbf{v}^* = \mathbf{0}$

QED



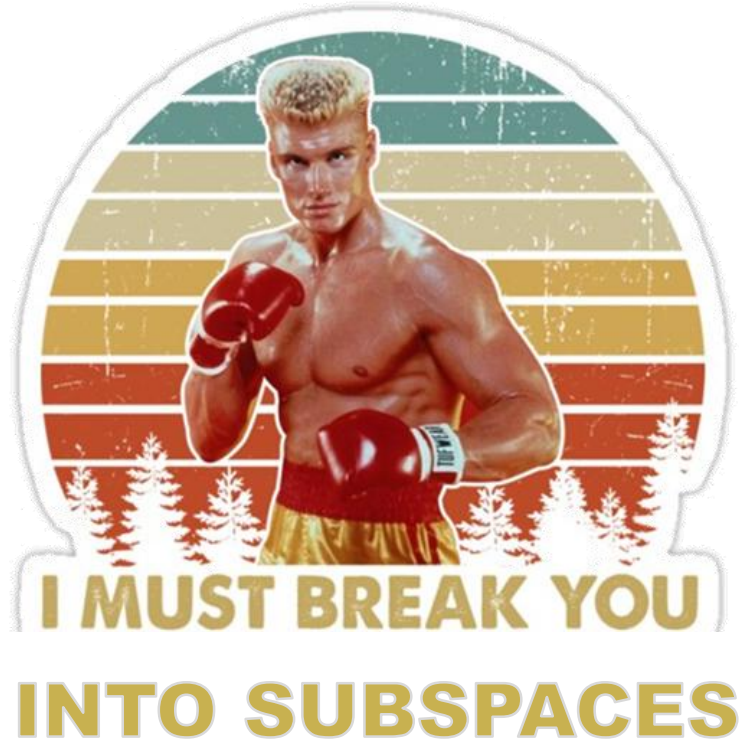


**Domain**

largest invertible subfunction

# Extracting an invertible Function (IX)

$$\underline{f : \mathcal{V} \longrightarrow \mathcal{W}}$$



To show about original function  $f$ :

$$\text{original domain } \underline{\mathcal{V} = \text{Ker } f \oplus \mathcal{V}^*}$$

Vector Space

Vector Space  
largest invertible  
subfunction

This breakdown helps us understand the original function and the invertibility of that function



# Extracting an invertible Function - Example (I)

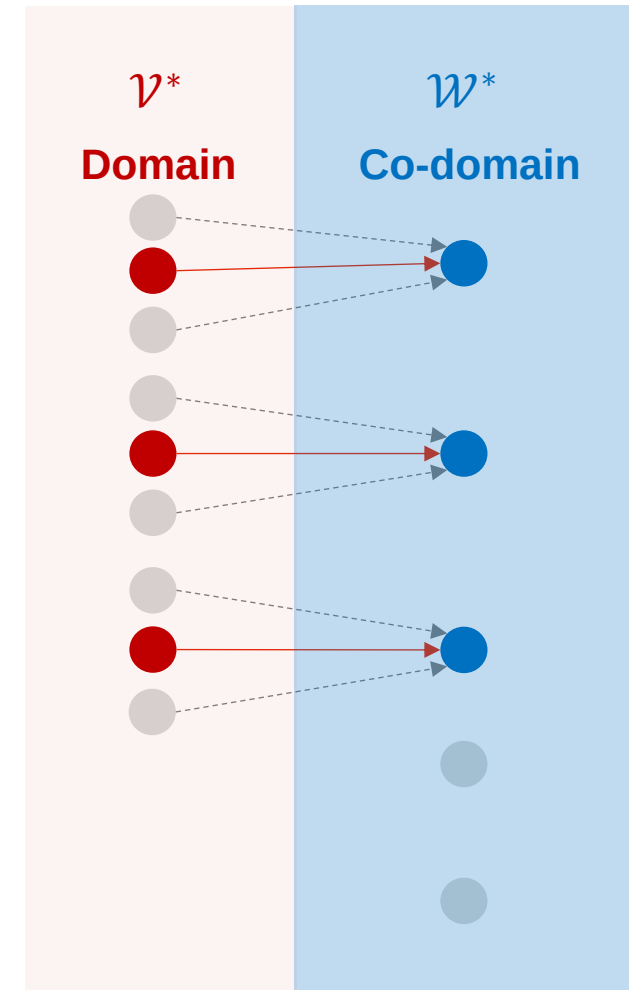
**Example:**

Let  $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$ , and define  $\mathbf{f} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$  by  $f(\mathbf{x}) = A\mathbf{x}$ .

Define  $\mathcal{W}^* = \text{Im } f = \text{Col } A = \text{Span } \{[1, 2, 1], [2, 1, 2], [1, 1, 1]\}$ .

We can now use the **Grow algorithm** (or e.g., **Gaussian elimination** [next week]) to find a basis for  $\mathcal{W}^*$

One basis for  $\mathcal{W}^*$  is  $\mathbf{w}_1 = [0, 1, 0]$ ,  $\mathbf{w}_2 = [1, 0, 1]$





# Extracting an invertible Function - Example (II)

Let  $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$ , and define  $\mathbf{f} : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$  by  $f(\mathbf{x}) = A\mathbf{x}$ .

One basis for  $\mathcal{W}^*$  is  $\mathbf{w}_1 = [0, 1, 0]$ ,  $\mathbf{w}_2 = [1, 0, 1]$

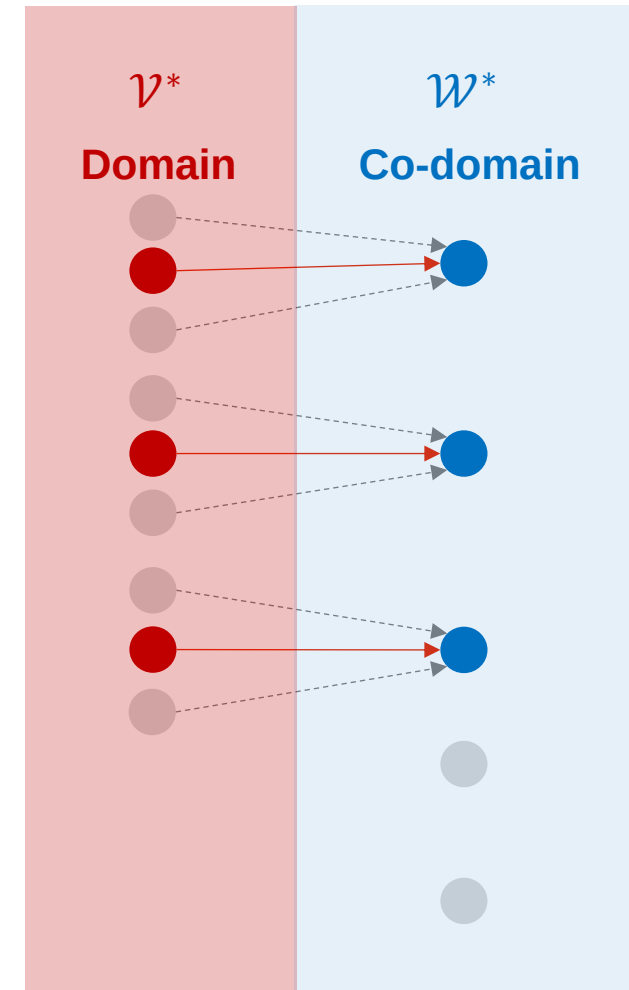
$A\mathbf{v}_1 = \mathbf{w}_1$  and  $A\mathbf{v}_2 = \mathbf{w}_2$ .

Using (e.g.,) a system of linear equations to solve

Pre-images for  $\mathbf{w}_1$  and  $\mathbf{w}_2$ :  $\mathbf{v}_1 = [\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}]$  and  $\mathbf{v}_2 = [-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ ,

Let  $\mathcal{V}^* = \text{Span} \{\mathbf{v}_1, \mathbf{v}_2\}$

Then  $f^* : \mathcal{V}^* \longrightarrow \text{Im } f$  is onto and one-to-one.





## Domain

# Extracting an invertible Function - Example (II)

original domain  $\mathcal{V} = \text{Ker } f \oplus \mathcal{V}^*$

$$f(\mathbf{x}) = A\mathbf{x}. \quad A = \left[ \begin{array}{c|c|c} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{array} \right]$$

$$\mathbf{v}_1 = \left[ \frac{1}{2}, -\frac{1}{2}, \frac{1}{2} \right] \text{ and } \mathbf{v}_2 = \left[ -\frac{1}{2}, \frac{1}{2}, \frac{1}{2} \right] \longrightarrow \mathcal{V}^* = \text{Span} \{ \mathbf{v}_1, \mathbf{v}_2 \}$$

By setting  $Ax = 0$  to find  $\text{Null } A$ :  $\text{Ker } f = \text{Span} \{ [1, 1, -3] \}$  [Null Space of A]

$$\text{Therefore } \mathcal{V} = \overbrace{(\text{Span} \{ [1, 1, -3] \}) \oplus (\text{Span} \{ \mathbf{v}_1, \mathbf{v}_2 \})}^{\text{Original Domain } \mathcal{V}}$$

$\underbrace{\hspace{10em}}_{\text{Ker } f}$ 
 $\underbrace{\hspace{10em}}_{\text{Smaller Domain } \mathcal{V}^*}$



## Domain

# Extracting an invertible Function - Example (IV)

**Vector Space:**

original domain  $\mathcal{V} = \text{Ker } f \oplus \mathcal{V}^*$

By Direct-Sum Dimension Corollary,

**Dimension:**

$$\dim \mathcal{V} = \dim \text{Ker } f + \dim \mathcal{V}^*$$

Since  $\mathbf{v}_1, \dots, \mathbf{v}_r$  form a basis for  $\mathcal{V}^*$ ,  $\dim \mathcal{V}^* = r = \dim \text{Im } f$

Alternative: We saw that after we selected a basis for  $\mathcal{W}^*$  which is onto, so  $\dim \mathcal{W}^* = \dim \text{Im } f$ . The pre-images for the basis vectors of  $\mathcal{W}^*$  formed a basis for  $\mathcal{V}^*$  which had the same dimension as the basis for  $\mathcal{W}^*$ . Thus,  $\dim \mathcal{V}^* = \dim \mathcal{W}^* = \dim \text{Im } f$

We have proved...

### Kernel-Image Theorem:

For any linear function  $f : \mathcal{V} \rightarrow \mathcal{W}$ ,

$$\dim \text{Ker } f + \dim \text{Im } f = \dim \mathcal{V}$$



# Linear Function Invertibility - Revisited

**Linear-Function Invertibility Theorem:** Let  $f : \mathcal{V} \longrightarrow \mathcal{W}$  be a linear function. Then  $f$  is invertible iff  $\dim \text{Ker } f = 0$  and  $\dim \mathcal{V} = \dim \mathcal{W}$ .



**Proof:** We saw before that  $f$

- ▶ is one-to-one iff  $\dim \text{Ker } f = 0$
- ▶ is onto if  $\dim \text{Im } f = \dim \mathcal{W}$

Therefore  $f$  is invertible if  $\dim \text{Ker } f = 0$  and  $\dim \text{Im } f = \dim \mathcal{W}$ .

Kernel-Image Theorem states  $\dim \text{Ker } f + \dim \text{Im } f = \dim \mathcal{V}$

Therefore

$$\dim \text{Ker } f = 0 \text{ and } \dim \text{Im } f = \dim \mathcal{W}$$

iff

$$\dim \text{Ker } f = 0 \text{ and } \dim \mathcal{V} = \dim \mathcal{W}$$

Can we now go to Matrix Invertibility to be more practical?





# Rank Nullity Theorem (I)

## Kernel-Image Theorem:

For any linear function  $f : \mathcal{V} \rightarrow W$ ,  $\dim \text{Ker } f + \dim \text{Im } f = \dim \mathcal{V}$

Remember  
(some slides  
ago)

Function    Matrix

Apply Kernel-Image Theorem to the function  $f(\mathbf{x}) = A\mathbf{x}$ :

- ▶  $\text{Ker } f = \text{Null } A \rightarrow$  *Null space* of a matrix  $A$  is  $\{\mathbf{u} : A * \mathbf{u} = \mathbf{0}\}$ .
- ▶  $\dim \text{Im } f = \dim \text{Col } A = \text{rank } A$

**Rank-Nullity Theorem:** For any  $n$ -column matrix  $A$ ,

$$\text{nullity } A + \text{rank } A = n$$

$\dim \text{Null } A$      $\dim \text{Col } A$     *No. Columns*





# Rank Nullity Theorem (II)

Null space of a matrix  $A$  is  $\{\mathbf{u} : A * \mathbf{u} = \mathbf{0}\}$ .

**Rank-Nullity Theorem:** For any  $n$ -column matrix  $A$ ,

$$\text{nullity } A + \text{rank } A = n$$

**Definition:** The *nullity* of matrix  $A$  is  $\dim \text{Null } A$

(**Linear Independence** [see proof on next slide])

If  $\text{Null } A$  only contains the zero vector (= is trivial)



then the *nullity*  $A$  is zero



$$\text{nullity } A + \text{rank } A = 0 + \text{rank } A = \text{rank } A$$



$$\text{rank } A = n$$

The dimension of the zero vector is 0



**Rank-Nullity Theorem:** For any  $n$ -column matrix  $A$ ,

$$\text{nullity } A + \text{rank } A = n$$

**Corollary:** Let  $A$  be an  $R \times C$  matrix. Then  $A$  is invertible if and only if  $|R| = |C|$  and the columns of  $A$  are linearly independent.

The function  $f$  is invertible    iff     $\dim \text{Ker } f = 0$  and  $\dim \mathbb{F}^C = \dim \mathbb{F}^R$   
                                                  iff    nullity  $A = 0$  and  $|C| = |R|$ .

## Square Matrix



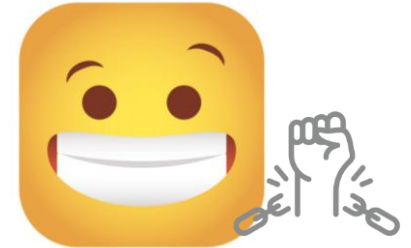


# Matrix Invertibility - Examples

$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$  is not square so cannot be invertible.

**Square Matrix** with  
**Linearly independent columns**

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$  is square and its columns are linearly independent so it is invertible.



$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix}$  is square but columns not linearly independent so it is not invertible.





# More on Matrix Invertibility

Any Matrix  
(incl. non-square)

Earlier we proved: *If  $A$  has an inverse  $A^{-1}$  then  $AA^{-1}$  is identity matrix*

**Converse:** If  $BA$  is identity matrix then  $A$  and  $B$  are inverses? **Not always true.**

It needed to be true both ways

**Corollary:**  $A$  and  $B$  are inverses of each other iff both  $AB$  and  $BA$  are identity matrices.



Two-ways

**Theorem:** Suppose  $A$  and  $B$  are square matrices such that  $BA$  is an identity matrix  $\mathbb{1}$ . Then  $A$  and  $B$  are inverses of each other.

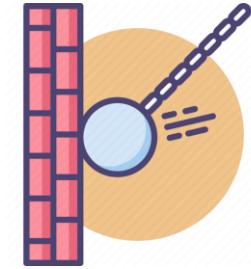
One-way

Square  
Matrix



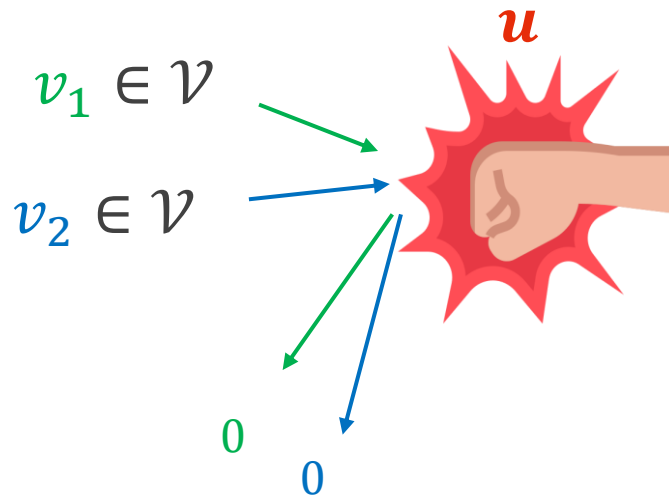


# Annihilator of a Vector Space (I)



**Definition:** For a subspace  $\mathcal{V}$  of  $\mathbb{F}^n$ , the *annihilator* of  $\mathcal{V}$ , written  $\mathcal{V}^\circ$ , is

$$\mathcal{V}^\circ = \{\mathbf{u} \in \mathbb{F}^n : \mathbf{u} \cdot \mathbf{v} = 0 \text{ for every vector } \mathbf{v} \in \mathcal{V}\}$$

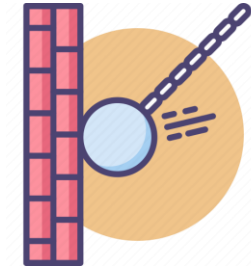


Now I am become the  
**Annihilator**, the  
Destroyer of **Vectors**





# Annihilator of a Vector Space (II)



One way to determine the Annihilator

- Which vectors  $\mathbf{u}$  satisfy  $\mathbf{u} \cdot \mathbf{v} = 0$  for every vector  $\mathbf{v}$  in  $\text{Span} \{[1, 0, 1], [0, 1, 0]\}$ ?

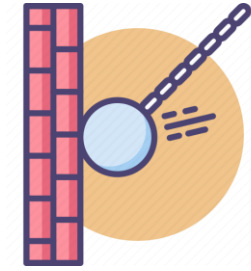
$$\mathbf{u} = [u_1, u_2, u_3]$$

$$\begin{aligned} [u_1, u_2, u_3] * [0, 1, 0] &= 0 \rightarrow u_2 = 0 \\ [u_1, u_2, u_3] * [1, 0, 1] &= 0 \rightarrow u_1 + u_3 = 0 \rightarrow u_3 = -u_1 \end{aligned} \quad \Rightarrow \quad \mathbf{u} = [u_1, 0, -u_1] = u_1[1, 0, -1]$$

Only scalar multiples of  $[1, 0, -1]$ .  $\rightarrow \text{Span}\{[1, 0, -1]\}$  incl.  $[0, 0, 0]$



# Annihilator of a Vector Space (III)



**Example over  $\mathbb{R}$ :** Let  $\mathcal{V} = \text{Span} \{[1, 0, 1], [0, 1, 0]\}$ . Then  $\mathcal{V}^\circ = \text{Span} \{[1, 0, -1]\}$ :

► Note that  $[1, 0, -1] \cdot [1, 0, 1] = 0$  and  $[1, 0, -1] \cdot [0, 1, 0] = 0$ .

► For any scalar  $\beta$ ,

$$\beta [1, 0, -1] \cdot \mathbf{v} = \beta ([1, 0, -1] \cdot \mathbf{v}) = 0$$

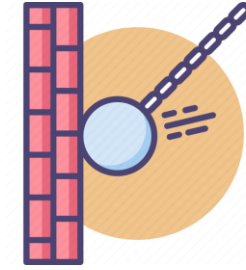
for every vector  $\mathbf{v}$  in  $\text{Span} \{[1, 0, 1], [0, 1, 0]\}$ .



$$\text{Span}\{[1, 0, -1]\} \subseteq \mathcal{V}^\circ$$



# Annihilator of a Vector Space (IV)



$$u * v = 0$$

$$u * (\alpha_1 v_1 + \alpha_2 v_2) = 0$$

$$\alpha_1 (u * v_1) + \alpha_2 (u * v_2) = 0$$

$$\alpha_1(0) + \alpha_2(0) = 0$$

We can annihilate anything  
the generators can create

Therefore  $[1, 0, -1] \cdot \mathbf{v} = 0$  for every vector  $\mathbf{v}$  in  $\text{Span} \{[1, 0, 1], [0, 1, 0]\}$ .

We have two linearly independent row vectors that are a basis for a matrix  $A$   
and we have  $\mathcal{V}^\circ$  which is the basis for the *Null A*

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

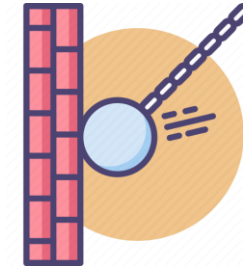
$$Ax = 0$$

$$\rightarrow \text{Null } A = \text{Span}([1 \ 0 \ -1])$$

Generator(s) for Null A



# Annihilator of a Vector Space (IV)



**Definition:** For a subspace  $\mathcal{V}$  of  $\mathbb{F}^n$ , the *annihilator* of  $\mathcal{V}$ , written  $\mathcal{V}^\circ$ , is

$$\mathcal{V}^\circ = \{\mathbf{u} \in \mathbb{F}^n : \mathbf{u} \cdot \mathbf{v} = 0 \text{ for every vector } \mathbf{v} \in \mathcal{V}\}$$

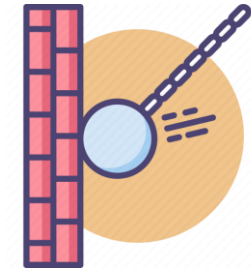
**Example over  $GF(2)$ :** Let  $\mathcal{V} = \text{Span} \{[1, 0, 1], [0, 1, 0]\}$ . Then  $\mathcal{V}^\circ = \text{Span} \{[1, 0, 1]\}$ :

- ▶ Note that  $[1, 0, 1] \cdot [1, 0, 1] = 0$  (remember  $GF(2)$  addition) and  $[1, 0, 1] \cdot [0, 1, 0] = 0$ .
- ▶ Therefore  $[1, 0, 1] \cdot \mathbf{v} = 0$  for every vector  $\mathbf{v}$  in  $\text{Span} \{[1, 0, 1], [0, 1, 0]\}$ .
- ▶ Of course  $[0, 0, 0] \cdot \mathbf{v} = 0$  for every vector  $\mathbf{v}$  in  $\text{Span} \{[1, 0, 1], [0, 1, 0]\}$ .
- ▶  $[1, 0, 1]$  and  $[0, 0, 0]$  are the only such vectors.  $= \text{Span}\{[1, 0, 1]\} \subseteq \mathcal{V}^\circ$  [in  $GF(2)$ ]





# Annihilator of a Vector Space (V)



**Example over  $\mathbb{R}$ :** Let  $\mathcal{V} = \text{Span} \{[1, 0, 1], [0, 1, 0]\}$ . Then  $\mathcal{V}^\circ = \text{Span} \{[1, 0, -1]\}$   
 $\dim \mathcal{V} + \dim \mathcal{V}^\circ = 3$

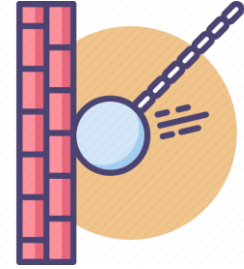
**Example over  $GF(2)$ :** Let  $\mathcal{V} = \text{Span} \{[1, 0, 1], [0, 1, 0]\}$ . Then  $\mathcal{V}^\circ = \text{Span} \{[1, 0, 1]\}$ .  
 $\dim \mathcal{V} + \dim \mathcal{V}^\circ = 3$

**Example over  $\mathbb{R}$ :** Let  $\mathcal{V} = \text{Span} \{[1, 0, 1, 0], [0, 1, 0, 1]\}$ .  
 Then  $\mathcal{V}^\circ = \text{Span} \{[1, 0, -1, 0], [0, 1, 0, -1]\}$ .  
 $\dim \mathcal{V} + \dim \mathcal{V}^\circ = 4$

**Annihilator Dimension Theorem:**  $\dim \mathcal{V} + \dim \mathcal{V}^\circ = n$



# Annihilator of a Vector Space (V)



**Annihilator Dimension Theorem:**  $\dim \mathcal{V} + \dim \mathcal{V}^o = n$



**Proof:** Let  $\mathbf{a}_1, \dots, \mathbf{a}_m$  be generators for  $\mathcal{V}$ .

$$\text{Let } A = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix} \text{ Then } \mathcal{V}^o = \text{Null } A.$$

Rank-Nullity Theorem states that

$$\begin{array}{rclcl} \text{rank } A & + & \text{nullity } A & = & n \\ \dim \mathcal{V} & + & \dim \mathcal{V}^o & = & n \end{array} \begin{array}{l} \text{(Vector Spaces)} \\ \text{(Matrix)} \end{array}$$

QED



# Exercise: Invertible Function & Annihilator

You are given the following matrix  $A$  for the function  $f(\mathbf{x}) = A\mathbf{x}$  :

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

- 1) Using the column vectors in matrix  $A$  we can define  $\mathcal{W}^* = \text{Span}(\{[1, 2, 0], [3, 2, 4], [1, 1, 1]\})$ . Show whether the three vectors are linearly independent.
- 2) Assume that one basis for  $\mathcal{W}^*$  is  $w_1 = [1, 0, 2], w_2 = [0, 1, -1]$ . What is one set of basis vectors for  $\mathcal{V}^*$ ?
- 3) What is the rank of the column vectors in  $A$ ? What is the nullity  $A$ ?
- 4) What would you proceed to determine the annihilator for the column space of  $A$ ? Explain.



## Solution

# Exercise: Invertible Function & Annihilator

- a) To determine whether  $a_1$ ,  $a_2$ , and  $a_3$  are **linearly independent**, we need to show that the zero vector can **only** be written via a trivial linear combination (= all alpha coefficients zero) of these vectors. So,

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

$$\mathbf{0} = \alpha_1 a_1 + \alpha_2 a_2 + \alpha_3 a_3 \quad \text{only with} \quad \alpha_1, \alpha_2, \alpha_3 = 0$$

$$[0, 0, 0] = \alpha_1 [1, 2, 0] + \alpha_2 [3, 2, 4] + \alpha_3 [1, 1, 1]$$

$$= [\alpha_1 + 3\alpha_2 + \alpha_3, 2\alpha_1 + 2\alpha_2 + \alpha_3, 4\alpha_2 + \alpha_3]$$

Set of linear equations to solve

$$\begin{array}{c|c|c|c|c} \begin{array}{l} \alpha_1 + 3\alpha_2 + \alpha_3 = 0 \\ 2\alpha_1 + 2\alpha_2 + \alpha_3 = 0 \\ 4\alpha_2 + \alpha_3 = 0 \end{array} & \longleftrightarrow & \begin{array}{l} \alpha_1 + 3\alpha_2 + \alpha_3 = 0 \\ -4\alpha_2 - \alpha_3 = 0 \\ 4\alpha_2 + \alpha_3 = 0 \end{array} & \longleftrightarrow & \begin{array}{l} \alpha_1 + 3\alpha_2 + (-4\alpha_2) = 0 \\ \alpha_3 = -4\alpha_2 \\ 0 = 0 \end{array} & \longleftrightarrow & \begin{array}{l} \alpha_1 = \alpha_2 \\ \alpha_3 = -4\alpha_2 \\ 0 = 0 \end{array} & \begin{array}{l} [\alpha_2, \alpha_2, -4\alpha_2] \\ = \alpha_2 [1, 1, -4] \end{array} \\ R_2 - 2R_1 & & R_3 + R_2 & & & & \end{array}$$



## Solution

## Exercise: Invertible Function &amp; Annihilator

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

The vectors are not linearly independent as any multiple of  $[1, 1, -4]$  for the alpha coefficients leads to the zero vector (thus,  $\text{Null } A = \text{Span}([1, 1, -4])$ )

b) Since the basis vectors are  $w_1 = [1, 0, 2]$ ,  $w_2 = [0, 1, -1]$  we need to find  $v_1, v_2$  so that  $Av_1 = w_1$  and  $Av_2 = w_2$

$$[\alpha_1 + 3\alpha_2 + \alpha_3, 2\alpha_1 + 2\alpha_2 + \alpha_3, 4\alpha_2 + \alpha_3] = [1, 0, 2]$$

Solving the corresponding system of LE leads to:

$$[\alpha_2 - 1, \alpha_2, -4\alpha_2 + 2]$$

If e.g.,  $\alpha_2 = 1$  then  $v_1 = [0, 1, -2]$

$$[\alpha_1 + 3\alpha_2 + \alpha_3, 2\alpha_1 + 2\alpha_2 + \alpha_3, 4\alpha_2 + \alpha_3] = [0, 1, -1]$$

Solving the corresponding system of LE leads to:

$$[\alpha_2 + 1, \alpha_2, -4\alpha_2 - 1]$$

If e.g.,  $\alpha_2 = 1$  then  $v_2 = [2, 1, -5]$

Thus, one set of basis vectors for  $\mathcal{V}^*$  is  $v_1 = [0, 1, -2]$ ,  $v_2 = [2, 1, -5]$



## Solution

# Exercise: Invertible Function & Annihilator

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 2 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

- c) Since the basis for  $\mathcal{W}^*$  contains two basis vectors, the rank for  $A$  is 2. Since

$$\text{nullity } A + \text{rank } A = n$$

And we know that the number of columns in  $A$  is three, we know that  $\text{nullity } A$  is 1. Alternatively, we could have also known this from the fact that  $\text{Null } A$  is spanned by one vector (which is not the zero vector) and is, thus, of dimension 1.

- d) **Definition:** For a subspace  $\mathcal{V}$  of  $\mathbb{F}^n$ , the *annihilator* of  $\mathcal{V}$ , written  $\mathcal{V}^\circ$ , is

$$\mathcal{V}^\circ = \{\mathbf{u} \in \mathbb{F}^n : \mathbf{u} \cdot \mathbf{v} = 0 \text{ for every vector } \mathbf{v} \in \mathcal{V}\}$$

So  $uA = \mathbf{0}$  (zero vector) meaning that each column vector in  $A$  multiplied by  $u$  needs to equal 0. Since  $A$  is known, we could simply formulate the corresponding system of linear equations (only difference is that the first element in the multiplication is unknown, not the second as usually) and solve for  $u$ .



# Some Key Take-aways (Lecture 14)

If two vector spaces  $\mathcal{U}$  and  $\mathcal{V}$  only share the zero vector, then we can define the **direct sum** of  $\mathcal{U}$  and  $\mathcal{V}$  as  $\mathcal{U} \oplus \mathcal{V}$ , which is also a vector space.

The union of a basis of  $\mathcal{U}$  and a basis of  $\mathcal{V}$  is a basis of  $\mathcal{U} \oplus \mathcal{V}$ . *[Direct Sum Basis Lemma]* and  $\dim \mathcal{U} + \dim \mathcal{V} = \dim \mathcal{U} \oplus \mathcal{V}$  *[Direct Sum Dimension Corollary]*

A **linear function is invertible** iff the kernel is trivial ( $\dim \text{Ker } f = 0$ ) [one-to-one] and the image of function  $f$  equals the co-domain  $\mathcal{W}$  ( $\dim \mathcal{V} = \dim \mathcal{W}$ ) [onto] *[Linear-Function Invertibility Theorem]*

We are using the dimension

For any  $n$ -column matrix  $A$ ,  $n = \text{rank } A + \text{nullity } A$  where  $\text{nullity } A$  refers to the dimension of the Null Space of  $A$  *[Rank-Nullity Theorem]*

Linear independence  
 $\rightarrow n = \text{rank } A$

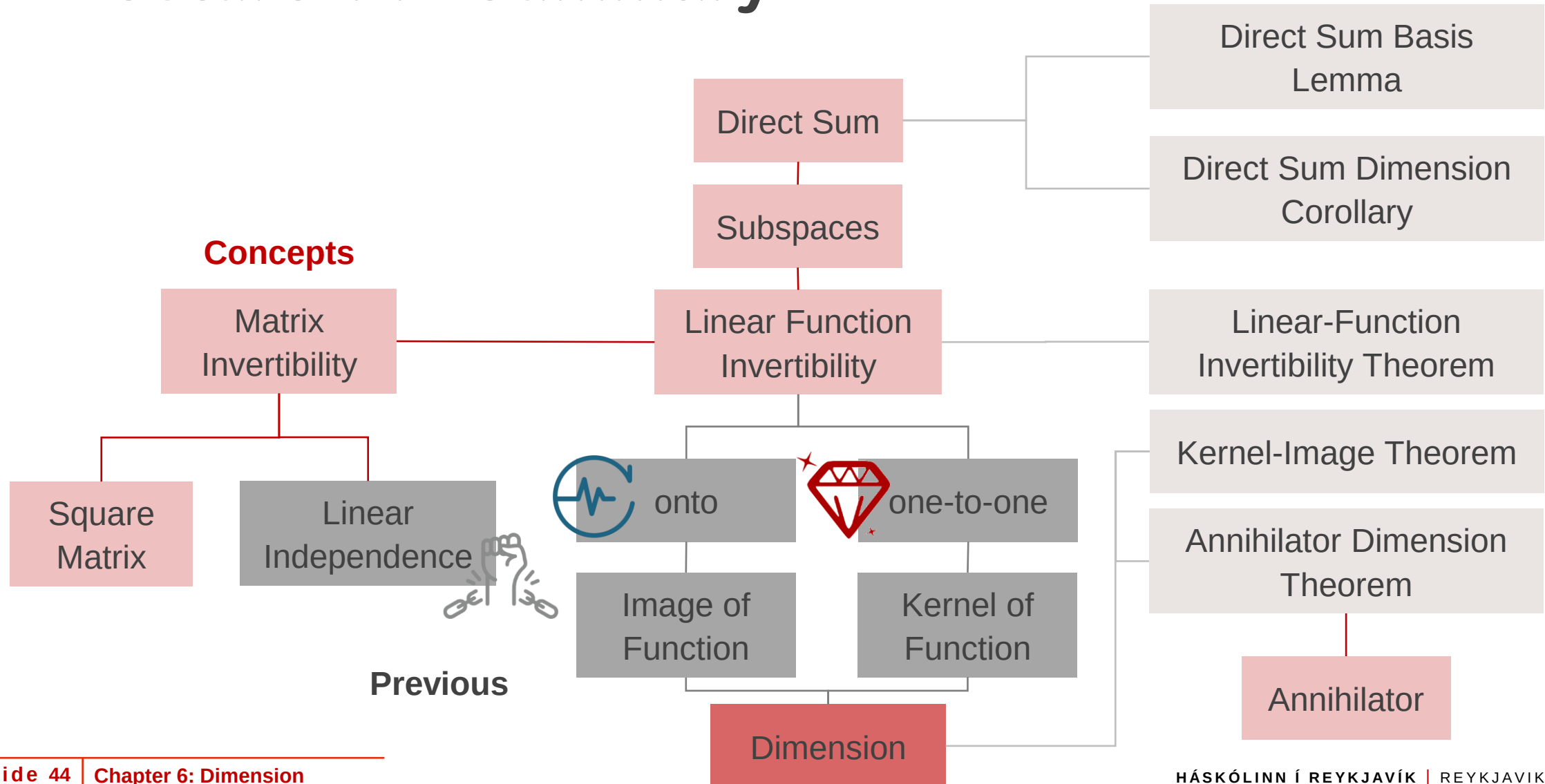
A **matrix is invertible** iff it is a square matrix and its columns are linearly independent

Very Practical!

The **annihilator** of  $\mathcal{V}$ , written  $\mathcal{V}^\circ$  is a set of vectors  $u$  whose dot product  $u * v$  is zero for every vector  $v \in \mathcal{V}$



# Lecture 14 - Summary





# Thank You For Your Attention!



**Accomplishment unlocked:  
Lecture 14 Completed**



# APPENDIX



# Cliffhanger - Function and Matrix Invertibility

**Question:** How can we tell if a matrix  $M$  is invertible?

From Lecture 10 (Chapter 4)

**Partial Answer:** By definition,  $M$  is an invertible matrix if the function  $f(\mathbf{x}) = M\mathbf{x}$  is an invertible function, i.e. if the function is one-to-one and onto.

- ✔ **One-to-one:** Since the function is linear, we know by the One-to-One Lemma that the function is one-to-one if its kernel is trivial, i.e. if the null space of  $M$  is trivial.
- ❓ **Onto:** We haven't yet answered the question *how we can tell if a linear function is onto?*

If we knew how to tell if a linear function is onto, therefore, we would know how to tell if a matrix is invertible.

**This lecture's journey is to be able to tell (1) if a linear function is invertible (specifically onto) and (2) if a matrix is invertible**

Don't leave me hanging!





# Direct Sum (III) – Alternative Version (Book)

Vectors over  $\mathbb{R}$ :

**Example:** Let  $\mathcal{U} = \text{Span} \{[1, 2, 1, 2], [3, 0, 0, 4]\}$  and let  $\mathcal{V}$  be the null space of

$$\begin{bmatrix} 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}.$$

- ▶ The vector  $[2, -2, -1, 2]$  is in  $\mathcal{U}$  because it is  $[3, 0, 0, 4] - [1, 2, 1, 2]$
- ▶ It is also in  $\mathcal{V}$  because

$$\begin{bmatrix} 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Therefore we cannot form  $\mathcal{U} \oplus \mathcal{V}$ .

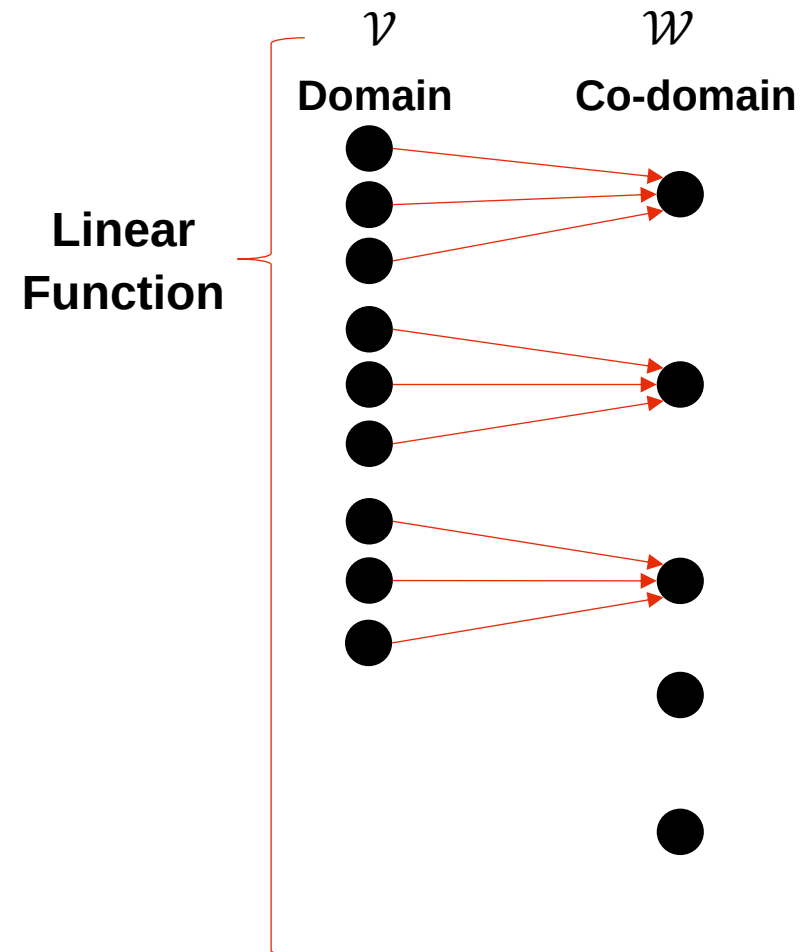


# Extracting an invertible Function (I)

Starting with a linear function  $f$  we will extract a largest possible subfunction that is invertible.

Make it **onto** by setting co-domain to be image of  $f$ .

Make it **one-to-one** by getting rid of extra domain elements sharing same image.



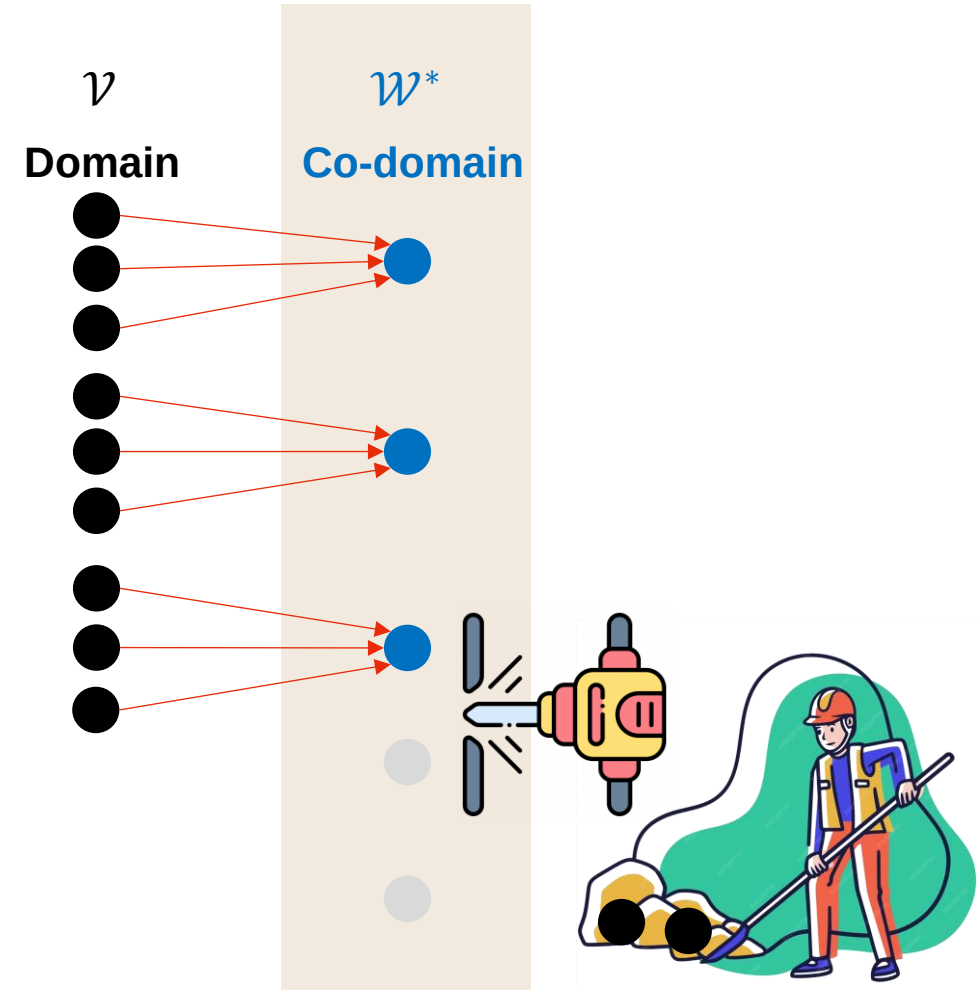


# Extracting an invertible Function (II)

Starting with a linear function  $f$  we will extract a largest possible subfunction that is invertible.

Start with linear function  $f : \mathcal{V} \longrightarrow \mathcal{W}$

**Step 1:** Choose smaller co-domain  $\mathcal{W}^*$





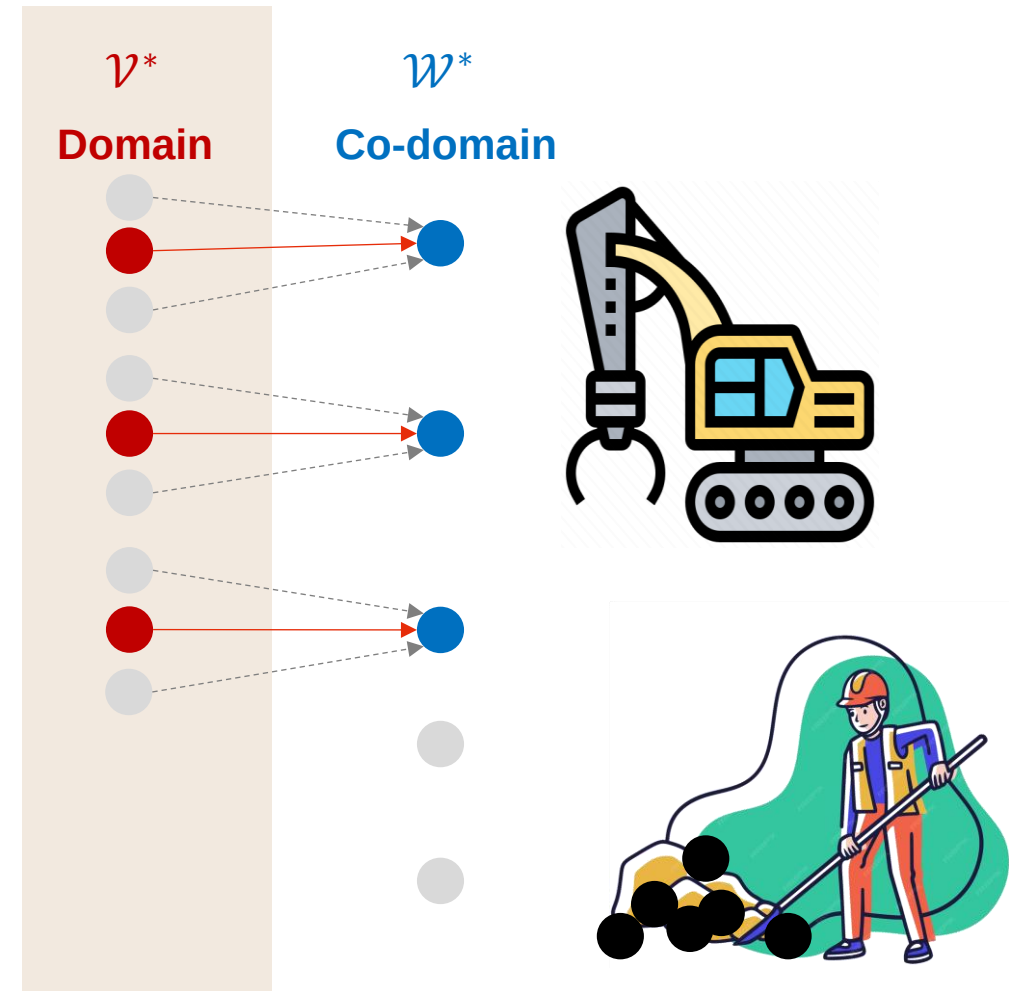
# Extracting an invertible Function (III)

Starting with a linear function  $f$  we will extract a largest possible subfunction that is invertible.

Start with linear function  $f : \mathcal{V} \longrightarrow \mathcal{W}$

Step 1: Choose smaller co-domain  $\mathcal{W}^*$

Step 2: Choose smaller domain  $\mathcal{V}^*$





# Extracting an invertible Function (IV)

Starting with a linear function  $f$  we will extract a largest possible subfunction that is invertible.

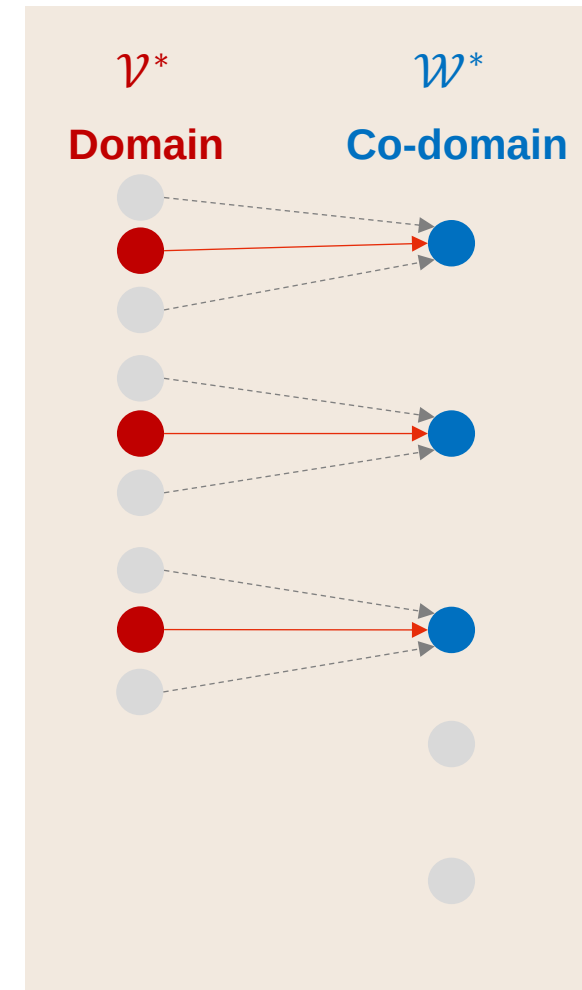
Start with linear function  $f : \mathcal{V} \longrightarrow \mathcal{W}$

Step 1: Choose smaller co-domain  $\mathcal{W}^*$

Step 2: Choose smaller domain  $\mathcal{V}^*$

Step 3: Define function  $f^* : \mathcal{V}^* \longrightarrow \mathcal{W}^*$  by  
 $f^*(\mathbf{x}) = f(\mathbf{x})$

In fact, we will end up selecting a *basis* of  $\mathcal{W}^*$  and a basis of  $\mathcal{V}^*$ .



After our job is done, we have an invertible function!



## Domain

## Extracting an invertible Function (X)

$$\underline{f : \mathcal{V} \longrightarrow \mathcal{W}}$$

$$\underline{\mathcal{V} = \text{Ker } f \oplus \mathcal{V}^*}$$

Must prove two things:

So, we break up vector space  $\mathcal{V}$  into two vector spaces

1.  $\text{Ker } f$  and  $\mathcal{V}^*$  share only zero vector (requirement for direct sum)

2. every vector in  $\mathcal{V}$  is the sum of a vector in  $\text{Ker } f$  and a vector in  $\mathcal{V}^*$

(Basis)

We already showed kernel of  $f^*$  is trivial. This shows only vector of  $\text{Ker } f$  in  $\mathcal{V}^*$  is zero vector. ( $f^*$  is one-to-one so  $\text{Ker } f^*$  is trivial  $\rightarrow$  only zero vector  $\rightarrow$  only one that can be in common)

Let  $\mathbf{v}$  be any vector in  $\mathcal{V}$ , and let  $\mathbf{w} = f(\mathbf{v})$ . Since  $f^*$  is onto, its domain  $\mathcal{V}^*$  contains a vector  $\mathbf{v}^*$  such that  $f(\mathbf{v}^*) = \mathbf{w}$ . Therefore  $f(\mathbf{v}) = f(\mathbf{v}^*)$  so  $f(\mathbf{v}) - f(\mathbf{v}^*) = \mathbf{0}$  so  $f(\mathbf{v} - \mathbf{v}^*) = \mathbf{0}$ . Thus  $\mathbf{u} = \mathbf{v} - \mathbf{v}^*$  is in  $\text{Ker } f$  and  $\mathbf{v} = \mathbf{u} + \mathbf{v}^*$

(Kernel)



# Extracting an invertible Function - Basis $V^*$



**Bonus:**  $\mathbf{v}_1, \dots, \mathbf{v}_r$  form a basis for  $V^*$

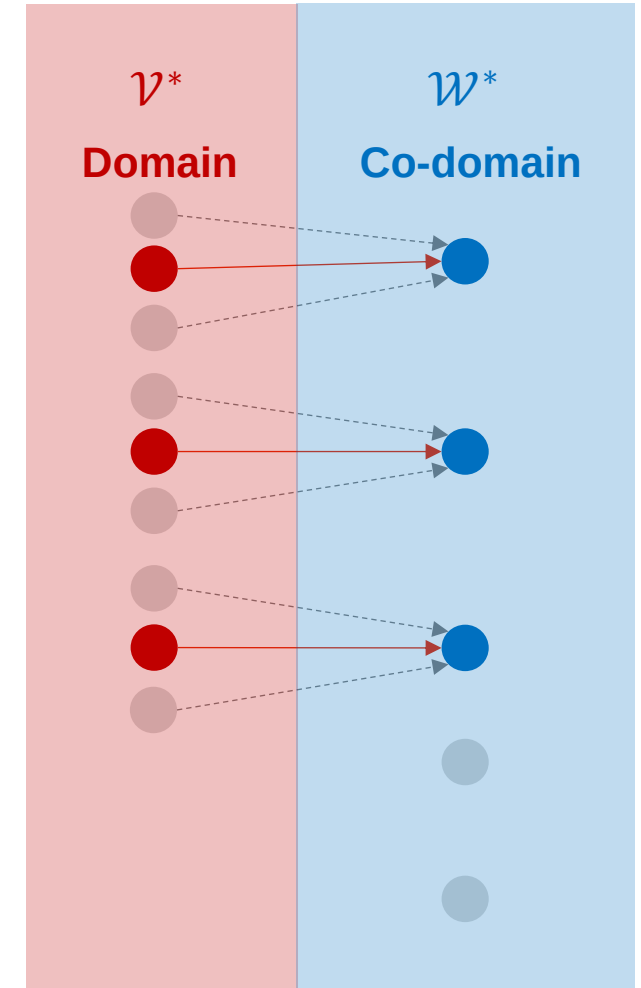
Need only show linear independence

Suppose  $\mathbf{0} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r$

Applying  $f$  to both sides,

$$\begin{aligned} \mathbf{0} &= f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_r \mathbf{v}_r) \\ &= \alpha_1 \mathbf{w}_1 + \dots + \alpha_r \mathbf{w}_r \end{aligned}$$

Because  $\mathbf{w}_1, \dots, \mathbf{w}_r$  are linearly independent,  $\alpha_1 = \dots = \alpha_r = 0$ . QED





# Invertibility of Transpose of Invertible Matrix

**Theorem:** The transpose of an invertible matrix is invertible.

$$A = \left[ \begin{array}{c|c|c} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{array} \right] = \left[ \begin{array}{c} \hline \mathbf{a}_1 \\ \vdots \\ \hline \mathbf{a}_n \end{array} \right] \quad A^T = \left[ \begin{array}{c|c|c} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{array} \right]$$

**Rank Theorem:** For every matrix  $M$ , row rank equals column rank.

Remember (last lecture)



**Proof:** Suppose  $A$  is invertible. Then  $A$  is **square** and its columns are **linearly independent**. Let  $n$  be the number of columns. Then  $\text{rank } A = n$ .

Because  $A$  is **square**, it has  $n$  rows. By Rank Theorem, rows are **linearly independent**.

Columns of transpose  $A^T$  are rows of  $A$ , so columns of  $A^T$  are **linearly independent**.

Since  $A^T$  is **square** and columns are **linearly independent**,  $A^T$  is invertible.

**QED**

If columns are, then rows are



# More Invertibility Matrix (II)

(assumed – see previous slide)

$BA$  is an identity matrix  $\mathbb{1}$ .



**Proof:** To show that  $A$  is invertible, need to show its columns are linearly independent

Let  $\mathbf{u}$  be any vector such that  $A\mathbf{u} = \mathbf{0}$ . Then  $B(A\mathbf{u}) = B\mathbf{0} = \mathbf{0}$ .

On the other hand,  $(BA)\mathbf{u} = \mathbb{1}\mathbf{u} = \mathbf{u}$ , so  $\mathbf{u} = \mathbf{0}$ . (see previous line)

**Null Space of  $A$  only contains the zero vector**

This shows  $A$  has an inverse  $A^{-1}$ . Now must show  $B = A^{-1}$ .

We know  $AA^{-1}$  is an identity matrix.

(what we want to prove)

$$BA = \mathbb{1}$$

$$(BA)A^{-1} = \mathbb{1}A^{-1}$$

by multiplying on the right by  $A^{-1}$

$$(BA)A^{-1} = A^{-1}$$

$$B(AA^{-1}) = A^{-1}$$

by associativity of matrix-matrix mult

$$B\mathbb{1} = A^{-1}$$

$$B = A^{-1}$$

**QED**



# Complementary Subspace (I)

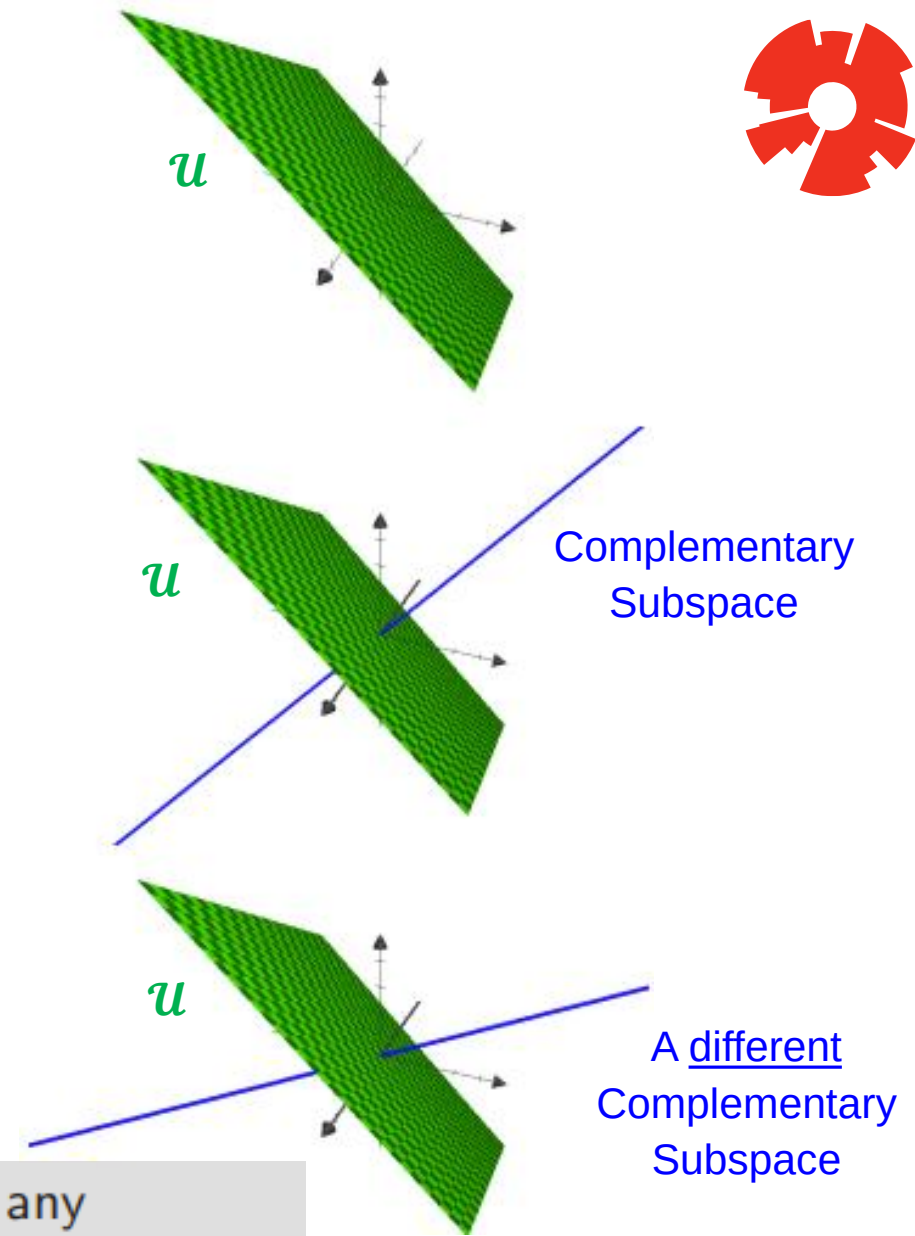
If  $\mathcal{U} \oplus \mathcal{V} = \mathcal{W}$ , we say  $\mathcal{U}$  and  $\mathcal{V}$  are complementary subspaces of  $\mathcal{W}$ .

**Example:** Suppose  $\mathcal{U}$  is a plane in  $\mathbb{R}^3$ .

Then any line through the origin that does not lie in  $\mathcal{U}$  is complementary subspace with respect to  $\mathbb{R}^3$

Example illustrates that, for a given subspace  $\mathcal{U}$  of  $\mathcal{W}$ , there can be many different subspaces  $\mathcal{V}$  such that  $\mathcal{U}$  and  $\mathcal{V}$  are complementary.

**Proposition:** For any finite-dimensional vector space  $\mathcal{W}$  and any subspace  $\mathcal{U}$ , there is a subspace  $\mathcal{V}$  such that  $\mathcal{U}$  and  $\mathcal{V}$  are complementary.





# Complementary Subspace (II) $\mathcal{U} \oplus \mathcal{V} = \mathcal{W}$ ,



**Proof:** Let  $\mathbf{u}_1, \dots, \mathbf{u}_k$  be a basis for  $\mathcal{U}$ . By Superset-Basis Lemma, there is a basis for  $\mathcal{W}$  that includes  $\mathbf{u}_1, \dots, \mathbf{u}_k$ :

$$B = \{\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{v}_1, \dots, \mathbf{v}_r\}$$

Let  $\mathcal{V} = \text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ .

Any vector in  $\mathcal{W}$  can be written in terms of its basis:

$$\mathbf{w} = \underbrace{\alpha_1 \mathbf{u}_1 + \dots + \alpha_k \mathbf{u}_k}_{\text{in } \mathcal{U}} + \underbrace{\beta_1 \mathbf{v}_1 + \dots + \beta_r \mathbf{v}_r}_{\text{in } \mathcal{V}}$$

If some vector  $\mathbf{v}$  is in  $\text{Span} \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$  and in  $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_r\}$  then  $\mathbf{v} = \alpha_1 \mathbf{u}_1 + \dots + \alpha_k \mathbf{u}_k$  and  $\mathbf{v} = \beta_1 \mathbf{v}_1 + \dots + \beta_r \mathbf{v}_r$

so

$$\alpha_1 \mathbf{u}_1 + \dots + \alpha_k \mathbf{u}_k = \beta_1 \mathbf{v}_1 + \dots + \beta_r \mathbf{v}_r$$

$$\mathbf{0} = \alpha_1 \mathbf{u}_1 + \dots + \alpha_k \mathbf{u}_k - \beta_1 \mathbf{v}_1 - \dots - \beta_r \mathbf{v}_r$$

so  $\alpha_1 = \dots = \alpha_k = \beta_1 = \dots = \beta_r = 0$  so  $\mathbf{v} = \mathbf{0}$ .

QED



(We discussed this problem previously in Lecture 6)

# Checksum Problem revisited

Checksum function maps  $n$ -vectors over  $GF(2)$  to 64-vectors over  $GF(2)$ :

$$\mathbf{x} \mapsto [\mathbf{a}_1 \cdot \mathbf{x}, \dots, \mathbf{a}_{64} \cdot \mathbf{x}]$$

Original “file”  $\mathbf{p}$ , transmission error  $\mathbf{e}$  so corrupted file is  $\mathbf{p} + \mathbf{e}$ .

If error is chosen according to uniform distribution, Probability ( $\mathbf{p} + \mathbf{e}$  has same checksum as  $\mathbf{p}$ ) =  $\frac{2^{\dim \mathcal{V}}}{2^n}$

where  $\mathcal{V}$  is the null space of the matrix

$$A = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_{64} \end{bmatrix}$$

**Fact:** Can easily choose  $\mathbf{a}_1, \dots, \mathbf{a}_{64}$  so that  $\text{rank } A = 64$   
(Randomly chosen vectors will probably work.)

**Rank-Nullity Theorem**  $\Rightarrow$

$$\text{rank } A + \text{nullity } A = n$$

$$64 + \dim \mathcal{V} = n$$

$$\dim \mathcal{V} = n - 64$$

Therefore  
Probability =  $\frac{2^{n-64}}{2^n} = \frac{1}{2^{64}}$

**very** tiny chance that the change is undetected



# Representation of Vector Spaces (I)

Two ways to represent a geometric object (line, plane, etc.) containing the origin:

- ▶ Span of some vectors
- ▶ Solution set of some system of linear equations with zero right-hand sides

Remember  
(Lecture 5 on  
Vector Spaces):

Two important ways to represent a vector space:

As the solution set of homogeneous linear system

$$\mathbf{a}_1 \cdot \mathbf{x} = 0, \dots, \mathbf{a}_m \cdot \mathbf{x} = 0$$

Equivalently,

$$\text{Null} \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}$$

As Span  $\{\mathbf{b}_1, \dots, \mathbf{b}_k\}$

Equivalently,

$$\text{Row} \begin{bmatrix} \mathbf{b}_1 \\ \vdots \\ \mathbf{b}_k \end{bmatrix}$$

How to transform between these two representations?



# Representation of Vector Spaces (II)

As the solution set of homogeneous linear system

$$\mathbf{a}_1 \cdot \mathbf{x} = 0, \dots, \mathbf{a}_m \cdot \mathbf{x} = 0$$

Equivalently,

$$\text{Null} \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}$$

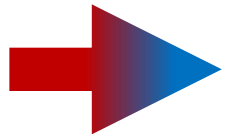
As Span  $\{\mathbf{b}_1, \dots, \mathbf{b}_k\}$

Equivalently,

$$\text{Row} \begin{bmatrix} \mathbf{b}_1 \\ \vdots \\ \mathbf{b}_k \end{bmatrix}$$

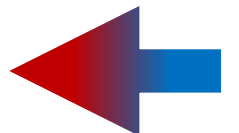
How to transform between these two representations?

**From left to right:** Given homogeneous linear system  $\mathbf{a}_1 \cdot \mathbf{x} = 0, \dots, \mathbf{a}_m \cdot \mathbf{x} = 0$ , find generators  $\mathbf{b}_1, \dots, \mathbf{b}_k$  for solution set



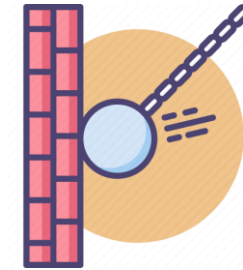
**From right to left:**

Given generators  $\mathbf{b}_1, \dots, \mathbf{b}_k$ , find homogeneous linear system  $\mathbf{a}_1 \cdot \mathbf{x} = 0, \dots, \mathbf{a}_m \cdot \mathbf{x} = 0$  whose solution set equals Span  $\{\mathbf{b}_1, \dots, \mathbf{b}_k\}$





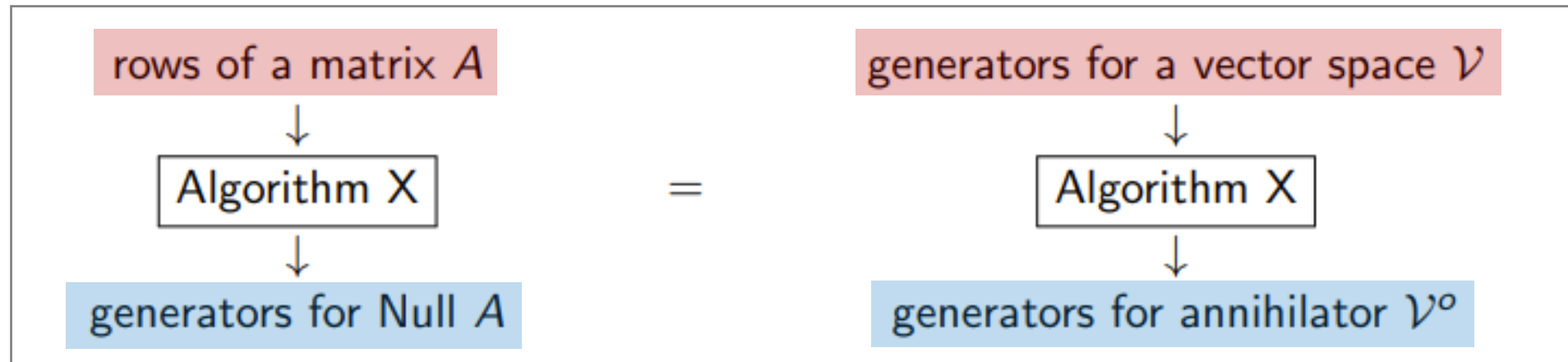
# Annihilator of a Vector Space (VI)



**Definition:** For a subspace  $\mathcal{V}$  of  $\mathbb{F}^n$ , the *annihilator* of  $\mathcal{V}$ , written  $\mathcal{V}^\circ$ , is

$$\mathcal{V}^\circ = \{\mathbf{u} \in \mathbb{F}^n : \mathbf{u} \cdot \mathbf{v} = 0 \text{ for every vector } \mathbf{v} \in \mathcal{V}\}$$

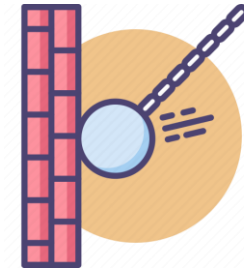
**From left to right:** Given system  $\mathbf{a}_1 \cdot \mathbf{x} = 0, \dots, \mathbf{a}_m \cdot \mathbf{x} = 0$ , find generators  $\mathbf{b}_1, \dots, \mathbf{b}_k$  for  
 ( Left ) ( Right ) for solution set



Algorithm X solves left-to-right problem.... what about right-to-left problem?

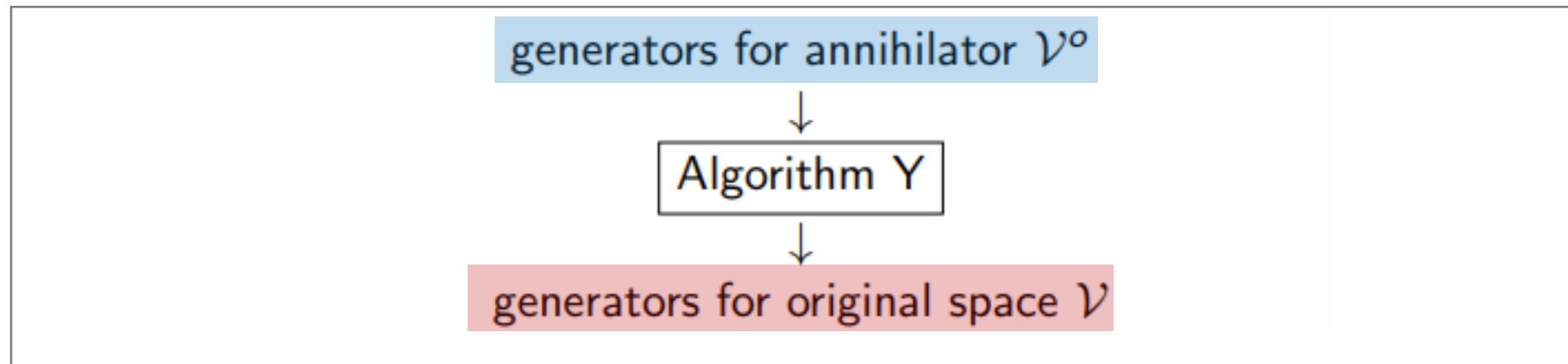


# Annihilator of a Vector Space (VII)



**From right to left:** Given generators  $\mathbf{b}_1, \dots, \mathbf{b}_k$ , find system  $\mathbf{a}_1 \cdot \mathbf{x} = 0, \dots, \mathbf{a}_m \cdot \mathbf{x} = 0$   
 ( Right ) whose solution set equals  $\text{Span} \{ \mathbf{b}_1, \dots, \mathbf{b}_k \}$

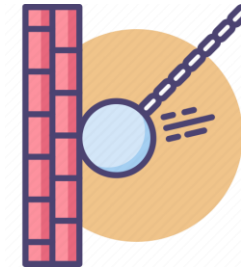
( Left )



**Theorem:**  $(\mathcal{V}^o)^o = \mathcal{V}$  (The annihilator of the annihilator is the original space.)



# Annihilator of a Vector Space (VIII)



**Theorem:**  $(\mathcal{V}^o)^o = \mathcal{V}$  (The annihilator of the annihilator is the original space.)

generators for a vector space  $\mathcal{V}$

Algorithm X

generators for annihilator  $\mathcal{V}^o$

Theorem shows:  
Algorithm X = Algorithm Y

What happens if we apply Algorithm X to generators for annihilator  $\mathcal{V}^o$ ?

generators for annihilator  $\mathcal{V}^o$

Algorithm X

generators for annihilator of annihilator  $(\mathcal{V}^o)^o$

=

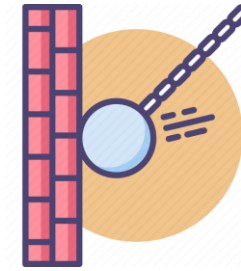
generators for annihilator  $\mathcal{V}^o$

Algorithm Y

generators for original space  $\mathcal{V}$



# Annihilator of a Vector Space (IX)



**Theorem:**  $(\mathcal{V}^\circ)^\circ = \mathcal{V}$  (The annihilator of the annihilator is the original space.)



## Proof:

Let  $\mathbf{a}_1, \dots, \mathbf{a}_m$  be a basis for  $\mathcal{V}$ . Let  $\mathbf{b}_1, \dots, \mathbf{b}_k$  be a basis for  $\mathcal{V}^\circ$ .

Since  $\mathbf{b}_1 \cdot \mathbf{v} = 0$  for every vector  $\mathbf{v}$  in  $\mathcal{V}$ ,

$$\mathbf{b}_1 \cdot \mathbf{a}_1 = 0, \mathbf{b}_1 \cdot \mathbf{a}_2 = 0, \dots, \mathbf{b}_1 \cdot \mathbf{a}_m = 0$$



Any  $b_i$  can annihilate any generator  $a$

Similarly  $\mathbf{b}_i \cdot \mathbf{a}_1 = 0, \mathbf{b}_i \cdot \mathbf{a}_2 = 0, \dots, \mathbf{b}_i \cdot \mathbf{a}_m = 0$  for  $i = 1, 2, \dots, k$ .

Reorganizing,

$$\mathbf{a}_1 \cdot \mathbf{b}_1 = 0, \mathbf{a}_1 \cdot \mathbf{b}_2 = 0, \dots, \mathbf{a}_1 \cdot \mathbf{b}_k = 0$$

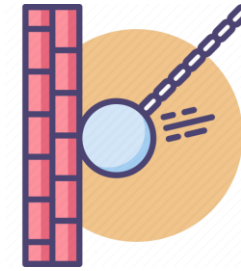
which implies that  $\mathbf{a}_1 \cdot \mathbf{u} = 0$  for every vector  $\mathbf{u}$  in  $\underbrace{\text{Span} \{ \mathbf{b}_1, \dots, \mathbf{b}_k \}}_{\mathcal{V}^\circ}$

$a_1$  can annihilate any vector  $u$

(continues next slide)



# Annihilator of a Vector Space (X)



**Theorem:**  $(\mathcal{V}^\circ)^\circ = \mathcal{V}$  (The annihilator of the annihilator is the original space.)



This shows  $\mathbf{a}_1$  is in  $(\mathcal{V}^\circ)^\circ$ . Similarly  $\mathbf{a}_2$  is in  $(\mathcal{V}^\circ)^\circ$ ,  $\mathbf{a}_3$  is in  $(\mathcal{V}^\circ)^\circ$ , ...,  $\mathbf{a}_m$  is in  $(\mathcal{V}^\circ)^\circ$ .

Therefore every vector in  $\text{Span} \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m\}$  is in  $(\mathcal{V}^\circ)^\circ$ .

Thus  $\underbrace{\text{Span} \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m\}}_{\mathcal{V}}$  is a subspace of  $(\mathcal{V}^\circ)^\circ$ .



All generators  $\mathbf{a}$   
can annihilate any  
annihilator  $u \in \mathcal{V}^\circ$

To show that these are equal, we must show that  $\dim \mathcal{V} = \dim(\mathcal{V}^\circ)^\circ$ .

By Annihilator Dimension Theorem,  $\dim \mathcal{V} + \dim \mathcal{V}^\circ = n$ .

By Annihilator Dimension Theorem applied to  $\mathcal{V}^\circ$ ,  $\dim \mathcal{V}^\circ + \dim(\mathcal{V}^\circ)^\circ = n$ .

Together these equations show  $\dim \mathcal{V} = \dim(\mathcal{V}^\circ)^\circ$ .

QED